

Assignment-2 (Due 09/05/2026)

Section 1. Multiple Choices. (For each question, please select the most appropriate answer.)

1.1. Which of the following complexes will be most reactive towards oxidative addition of H_2 ?

- (a) $\text{RhCl}(\text{CO})(\text{PPh}_3)_2$ (b) $\text{RhBr}(\text{CO})(\text{PPh}_3)_2$
 (c) $\text{IrCl}(\text{CO})(\text{PPh}_3)_2$ (d) $\text{IrBr}(\text{CO})(\text{PPh}_3)_2$

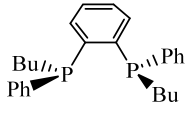
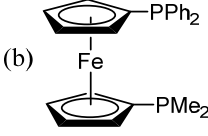
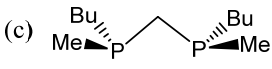
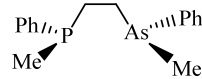
1.2. Which of the following oxidative addition reactions is/are not likely through a concerted mechanism?

- (a) $\text{IrI}(\text{CO})(\text{PPh}_3)_2 + \text{Me}_3\text{SiH} \longrightarrow \text{IrI}(\text{H})(\text{SiMe}_3)(\text{CO})(\text{PPh}_3)_2$
 (b) $\text{IrI}(\text{CO})(\text{PPh}_3)_2 + \text{MeCOCl} \longrightarrow \text{IrI}(\text{Cl})(\text{MeCO})(\text{CO})(\text{PPh}_3)_2$
 (c) $\text{IrI}(\text{CO})(\text{PPh}_3)_2 + \text{H}_2 \longrightarrow \text{IrI}(\text{H})_2(\text{CO})(\text{PPh}_3)_2$
 (d) $\text{IrCl}(\text{CO})(\text{PPh}_3)_2 + \text{HCl}$ (in aqueous solution) $\longrightarrow \text{IrHCl}_2(\text{CO})(\text{PPh}_3)_2$
 (e) two of them (f) three of them (g) all of them (h) none of them

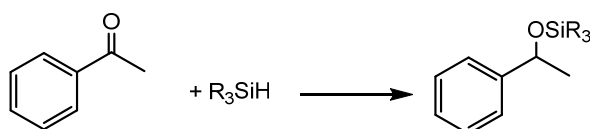
1.3. Which of the following reactions would you expect to have the largest equilibrium constant K_{eq} ?

- (a) $\text{Cp}(\text{CO})_3\text{Cr}-\text{CH}_3 + \text{CO} \rightleftharpoons \text{Cp}(\text{CO})_3\text{Cr}-\text{COCH}_3$
 (b) $\text{Cp}(\text{CO})_3\text{Mo}-\text{CH}_3 + \text{CO} \rightleftharpoons \text{Cp}(\text{CO})_3\text{Mo}-\text{COCH}_3$
 (c) $\text{Cp}(\text{CO})_3\text{W}-\text{CH}_3 + \text{CO} \rightleftharpoons \text{Cp}(\text{CO})_3\text{W}-\text{COCH}_3$
 (d) $\text{Cp}(\text{CO})_3\text{W}-\text{H} + \text{CO} \rightleftharpoons \text{Cp}(\text{CO})_3\text{W}-\text{COH}$

1.4. Which of the following ligands could be used for asymmetric catalytic hydrogenation of $\text{C}_6\text{H}_5\text{CH}=\text{CMeCONMe}_2$?

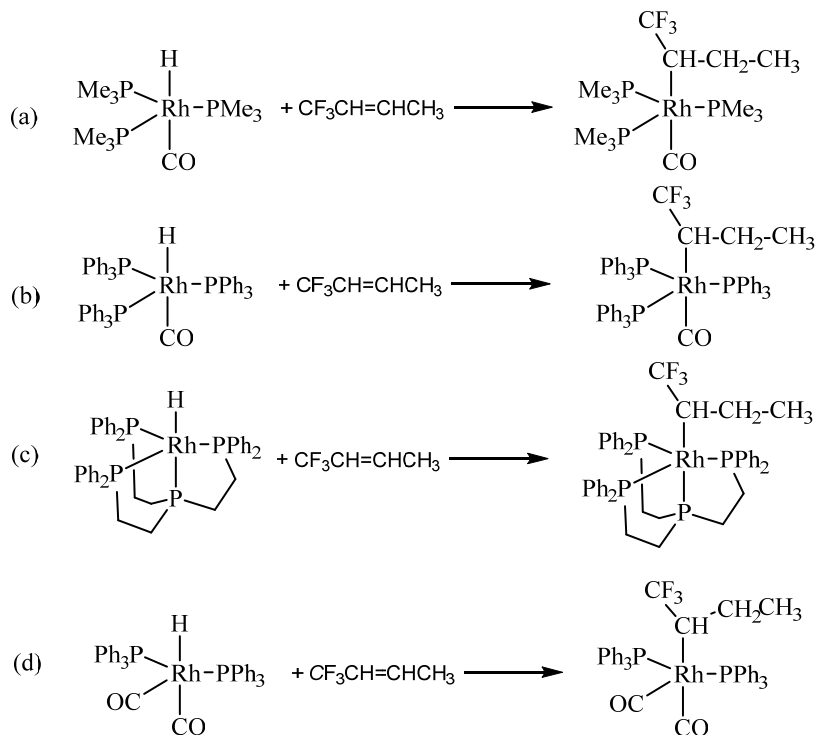
- (a)  (b)  (c) 
 (d) 
 (e) two of them (f) three of them (g) all of them (h) none of them

1.5. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?

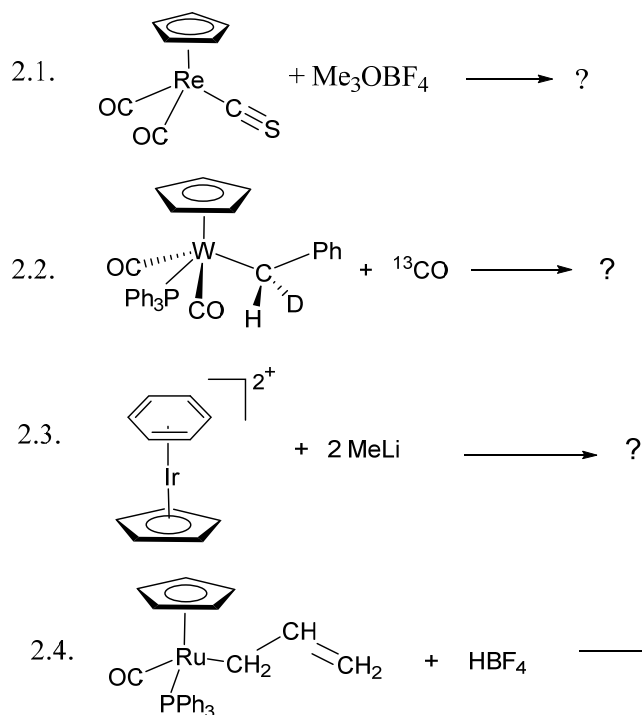


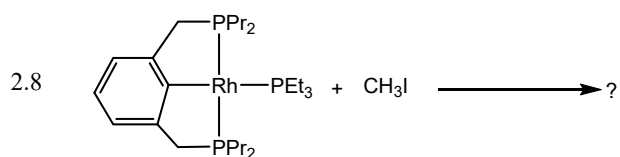
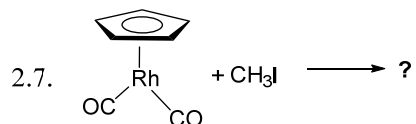
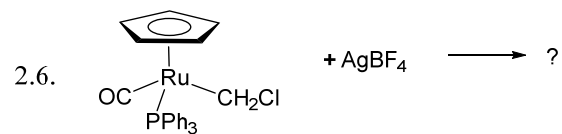
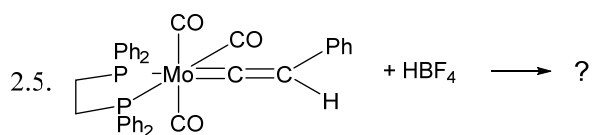
- (a) $\text{Pt}(\text{PPh}_3)_4$ (b) $\text{RhCl}(\text{PPh}_3)_3$ (c) $\text{PtHCl}(\text{PPh}_3)_2$ (d) SiH_4
 (e) two of them (f) three of them (g) all of them (h) none of them

1.6. Which of the following reactions would you expect to have the highest reaction rate?

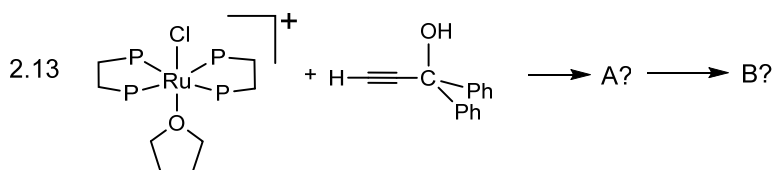
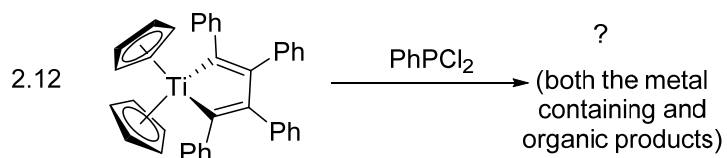
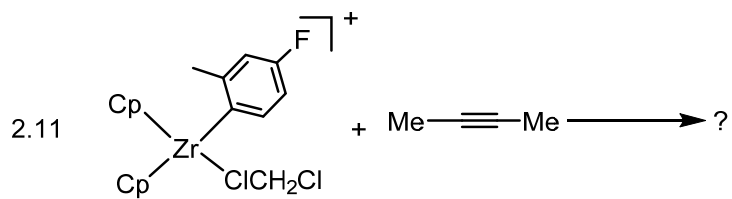
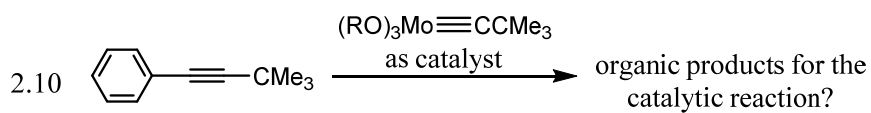
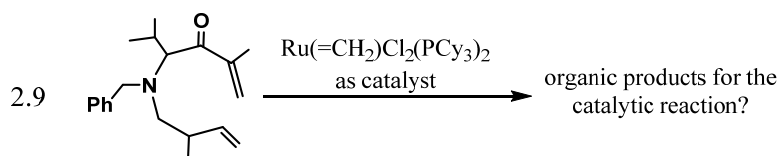


Section 2. Predict the products of the following reactions

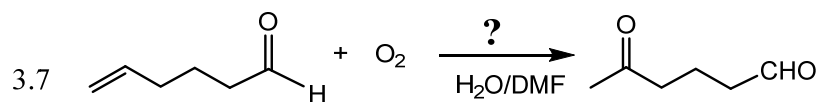
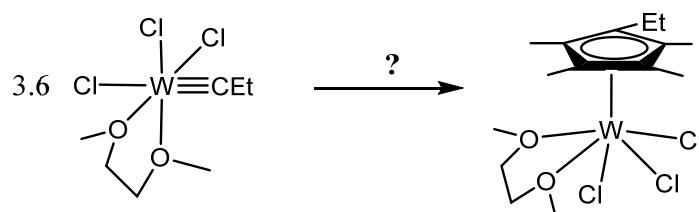
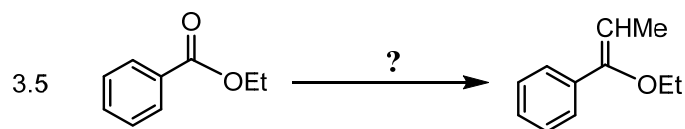
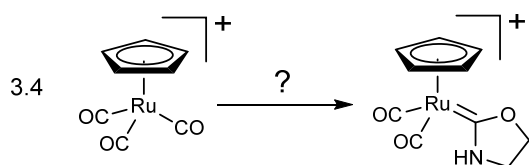
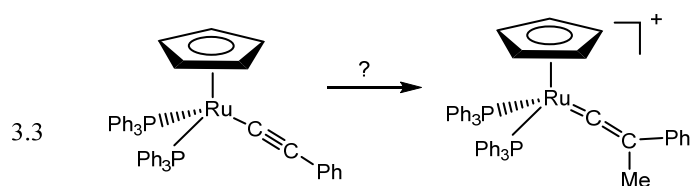
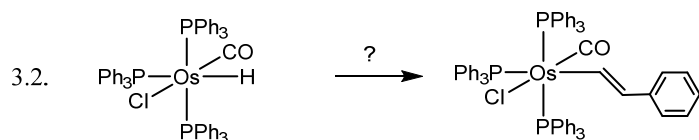
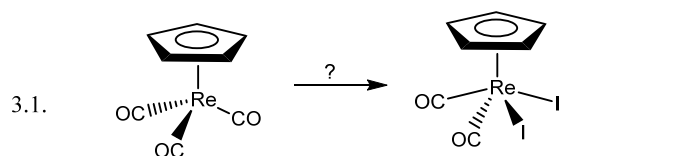


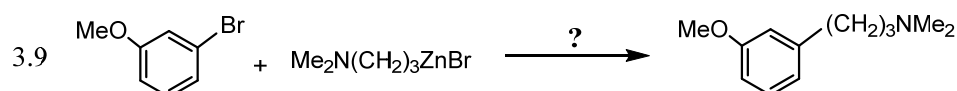
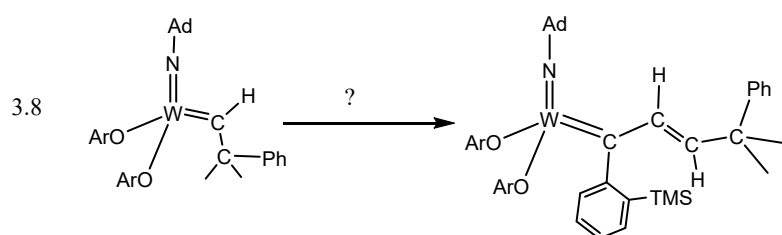


(Ref. *Organometallics*, **2009**, 28, 523)

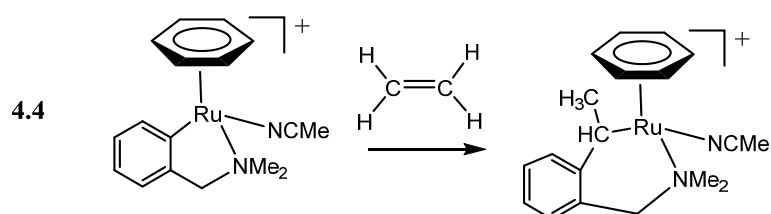
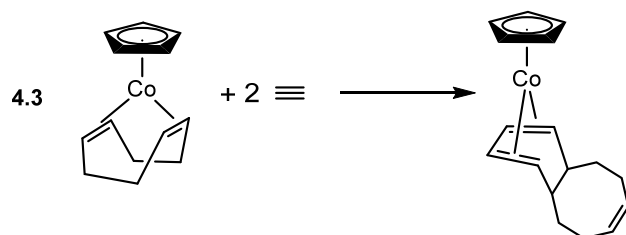
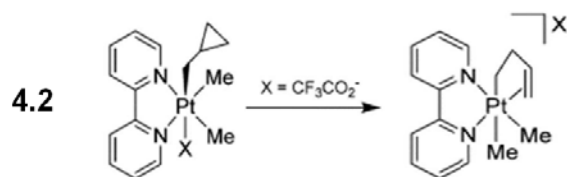
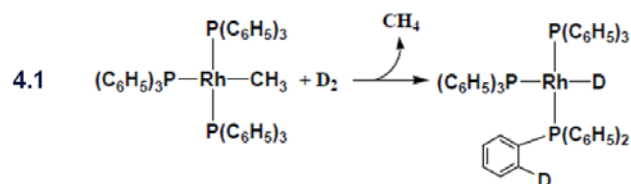


Section 3. Provide reagents for the following reactions (Note: Some of the reactions may need more than one reagents and/or catalyst(s).

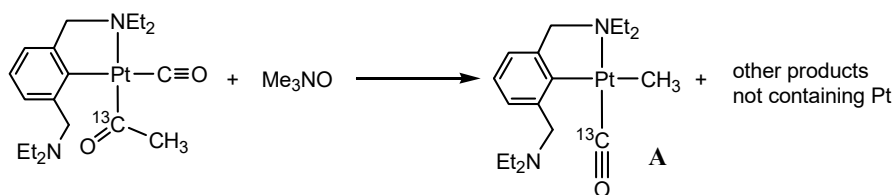




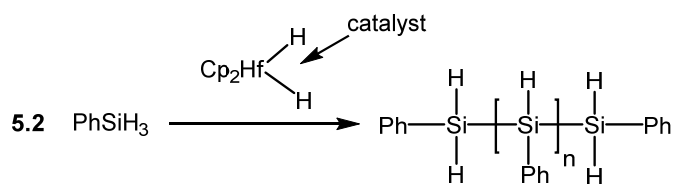
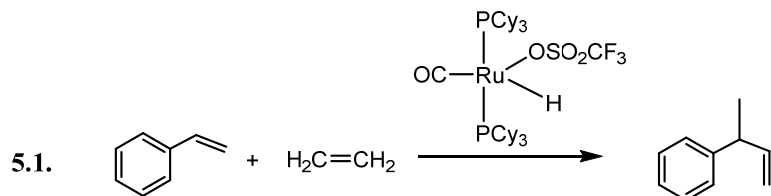
Section 4. Propose reasonable mechanisms for stoichiometric reactions.



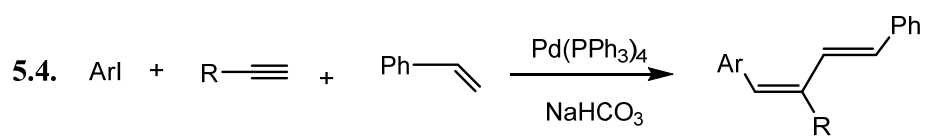
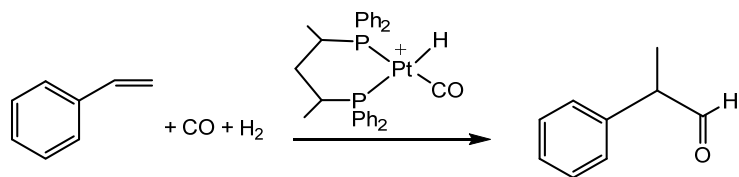
4.5 The following reaction was reported recently (see *Organometallics*, 2009, **28**, 1613). Propose a mechanism for the formation of product **A**. Be sure to indicate what other products are generated in the reaction.



Section 5. Propose a reasonable mechanism for the following catalytic reactions.



5.3 (Reference: *J. Am. Chem. Soc.* **2004**, 126, 5585)



(Optional)